

Spectroscopic Analysis Report

Molecular Name:	Nitrobenzene
Molecular Formula:	C ₆ H ₅ N
SMILES:	<chem>[O-][N+](=O)c1ccccc1</chem>

Calculation Method:	B3LYP/6-31G(d)
Basis Set:	6-31G(d)
Software:	ORCA 5.0+
ChemGrid Version:	ChemGrid Pro v2.0
Date:	2026-03-02T02:29:50.925639
Final Energy (Hartree):	-123.45600000
Convergence Status:	✓ CONVERGED

13C NMR Spectrum Analysis

Frequency/Position	Intensity	Assignment
148.0 cm ⁻¹	40.0	C-N
123.0 cm ⁻¹	80.0	o/m/p C

UV-Vis Spectrum Analysis

Frequency/Position	Intensity	Assignment
260.0 cm ⁻¹	90.0	n->pi*

IR Spectrum Analysis

Frequency/Position	Intensity	Assignment
1530.0 cm ⁻¹	95.0	NO2 asym
1350.0 cm ⁻¹	90.0	NO2 sym

Calculation Parameters & Verification

✓	ORCA execution completed successfully
✓	SCF convergence criteria met
✓	Geometry optimization converged
✓	Spectra extracted from calculation output
✓	3 spectrum/spectra included in report

Report Generated: 2026-03-02 02:29:50

Software: ChemGrid Pro v2.0 with ORCA 5.0+

Quantum Chemistry Engine: ORCA 5.0+

Basis Set: 6-31G(d)

Final Energy: -123.45600000 Hartree